Education and Employment Exits During COVID-19: Evidence from Brazil

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Abstract

We use the rotating panel of Brazil's *PNAD Contínua* to ask how the education gradient in the risk of leaving employment moved through the COVID-19 pandemic. Linking 3,024,050 workers across 52 quarters (2012Q1–2024Q4, 8,002,128 person-quarters), we estimate survey-weighted average predictive margins by education and quarter. Unconditionally the gradient never reverses: graduates leave employment at 4.0% per quarter, non-graduates at 10.7%. Compared within similar jobs, it does. For six consecutive quarters, from 2020Q2 to 2021Q3, graduates were 1.8 percentage points more likely to leave employment than otherwise similar non-graduates; the pre-pandemic gradient then returned. A decomposition attributes the unconditional advantage about equally, in every period, to where graduates work and to how they fare inside comparable jobs. Both channels compress proportionally during the window, so that gap halves without changing sign: graduates did not move into riskier jobs. The reversal emerges only once workers are also matched on pay, hours, tenure and contract type. It is concentrated among informal private employees, runs entirely through exits out of the labour force rather than into measured unemployment, and survives weighting by the inverse probability of being matched forward. The design is descriptive, but the episode shows that the education gradient in job security is not a structural constant: it moved in the segment of the labour market where a safety net segmented along the formal–informal margin was concentrated.

Keywords: Employment exits, Education, Informality, COVID-19, Brazil, Rotating panel

JEL classification: J63, J24, J46, I38, O17

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1. Introduction

Job security rises with schooling. Across the OECD, unemployment among workers with tertiary education averages 4–5%, against 12.8% for those without upper secondary education.[1] The COVID-19 recession initially widened that gradient almost everywhere: in the United States, unemployment among workers with at most a high school diploma rose by 11 percentage points between February and May 2020, three times the increase recorded for college graduates (Kochhar, 2020; Montenovo et al., 2022; Cajner et al., 2020), and the incidence of the shock was sharply unequal along skill lines more generally (Adams-Prassl et al., 2020; Albanesi and Kim, 2021; Aum et al., 2021). The same asymmetry shows up on the demand side, in vacancy postings and separations (Forsythe et al., 2020), and in high-frequency administrative data on employment losses by wage level (Chetty et al., 2024).

The mechanisms are familiar. Human capital raises productivity and lowers displacement risk (Ben-Porath, 1967; Mincer, 1974; Lise and Postel-Vinay, 2020); education sorts workers into more stable jobs and cushions them against cyclical downturns (Farber, 2005; Hoynes et al., 2012), into activities less exposed to reallocation shocks (Davis, 1987; Brainard and Cutler, 1993), and into tasks more amenable to remote work (Autor et al., 2003; Dingel and Neiman, 2020; Barrero et al., 2023). Causal evidence is scarcer, but Beuermann et al. (2024) exploit school-assignment discontinuities in Barbados and find that additional schooling did reduce the probability of losing a job after the COVID-19 onset, in a way their evidence attributes to skill rather than to signalling or to sectoral sorting.

This paper asks whether that gradient held in a labour market where two in five workers are informal and where the emergency policy response was explicitly segmented along the formal–informal margin. Informality on this scale is the norm rather than the exception in developing countries, and it is not a residual category: it shapes firm selection, wage setting and the response of employment to shocks (La Porta and Shleifer, 2014; Meghir et al., 2015; Ulyssea, 2018; Dix-Carneiro et al., 2026). Brazil is an informative case precisely because its two main interventions worked through different parts of the labour market. *Auxílio Emergencial*, a large cash transfer, paid monthly benefits from April 2020 through October 2021 and reached informal and low-income workers (Arruda et al., 2022; Neri, 2021; de Leon et al., 2023). In parallel, the *Programa Emergencial de Manutenção do Emprego e da Renda* let formal employers cut hours and wages or suspend contracts with partial income replacement (Ministério do Trabalho e Emprego, 2020). Because schooling and formality are strongly correlated (Ulyssea, 2018, 2020; Parente, 2024), the two programmes did not simply differ in generosity: they reached different education groups.

A substantial literature already documents the aggregate contraction of Brazilian and Latin American employment during COVID-19 and the disproportionate exposure of informal workers. Maurizio et al. (2023) show, for six Latin American countries including Brazil, that the fall in informal employment drove the

[1] OECD (2023), *Education at a Glance 2023.*

overall contraction, that it operated mainly through higher exit rates rather than lower entry, and that most informal workers who lost their jobs left the labour force altogether. Bottan et al. (2020) document unequal exposure across seventeen developing countries, and work on *Auxílio Emergencial* establishes both its coverage and its behavioural effects (Arruda et al., 2022; de Leon et al., 2023; Andrade and Souza, 2023). What this literature has not established is whether schooling itself continued to protect workers once the composition of jobs is held fixed. Nor, if it did not, whether that reflects a change in where graduates and non-graduates work or a change in the risk they face inside the same kind of job.

That is our contribution. We provide longitudinal evidence on quarterly employment exits by education over 2012Q1–2024Q4, and we decompose the education gap into a component reflecting where the two groups work and a component reflecting the risk they face inside comparable jobs, with particular attention to informal employment. Three features of the design deserve emphasis at the outset. We report adjusted exit probabilities for each education-by-quarter cell, as survey-weighted average predictive margins, and not only the contrasts between them: the education coefficients of a model with continuous covariates and high-dimensional fixed effects identify those contrasts but say nothing about either group's level. Inference is clustered two ways, by *PNAD Contínua* primary sampling unit and by year-quarter, and because we report 52 quarterly contrasts we accompany them with simultaneous rather than pointwise bands. And we take the matching of respondents across interviews seriously, documenting retention by education and formality and re-weighting by the inverse retention propensity.

The unconditional gradient is large and never reverses: graduates leave employment at 4.0% per quarter, non-graduates at 10.7%. Conditional on observables, however, the gradient does reverse. The adjusted gap is -0.009 before the pandemic, widens to -0.035 in 2020Q1 as non-graduates absorb the first shock, and turns positive from 2020Q2 through 2021Q3, peaking at 0.023 in 2020Q3, with graduates leaving employment more often than comparable non-graduates. It returns to -0.010 thereafter. Inference disciplines which of these movements we can lean on. The onset widening rests on a single quarter, and a wild cluster bootstrap on year-quarter cannot distinguish it from an idiosyncratic quarterly shock; the reversal rests on six quarters and survives the same test comfortably. We report the first and build on the second.

The decomposition then tells us what the reversal is not. We split the unconditional gap into a composition component, which captures graduates and non-graduates working in different formality, sector and occupation cells, and a within-cell component, which captures the difference in their exit rates inside a given cell. Composition takes roughly 55% of the gap and within-cell risk the remaining 45%, and those shares barely move across periods. During 2020Q2–2021Q3 both components compress at almost the same rate, composition by 45% and within-cell risk by 39%, so the unconditional gap halves without changing sign. Graduates did not reallocate into riskier cells. The reversal appears only once workers are matched on job characteristics as well as on cells, and it is concentrated among informal private employees.

We are explicit about what this does and does not establish. The design is descriptive. The timing of the reversal coincides with the emergency programmes, but we do not exploit variation in eligibility or exposure and therefore do not identify their effects. What the evidence does deliver is a fact that any account of the episode has to accommodate: for six quarters, in the segment of the Brazilian labour market that a broad cash transfer reached, schooling stopped predicting who kept working.

The remainder of the paper proceeds as follows. Section 2 describes the two emergency programmes. Section 3 describes the matched panel and the outcome. Section 4 sets out the specification, the margins and the inference. Section 5 presents the results, Section 6 the retention analysis and Section 7 the robustness checks. Section 8 interprets the findings and Section 9 concludes.

2. Institutional background

Brazil's emergency response combined a large cash transfer with a formal-sector job-preservation scheme, and the two instruments reached different workers.

Auxílio Emergencial began in April 2020 and ran, at declining generosity, until October 2021. It targeted informal workers, the self-employed, the unemployed and low-income households, initially paying R$600 per month, above the minimum wage for a household with two eligible adults, and later R$300 and then R$150–375 in the 2021 extension. At its peak it reached on the order of 68 million people, one of the largest emergency transfers implemented anywhere (Arruda et al., 2022; Neri, 2021; Salto et al., 2021). Evidence indicates it materially raised incomes at the bottom of the distribution and changed labour supply behaviour, including hours worked (de Leon et al., 2023).

The *Programa Emergencial de Manutenção do Emprego e da Renda*, established by Provisional Measure 936/2020 and converted into Law 14,020/2020, operated on the formal margin. It allowed firms to reduce hours and wages proportionally or to suspend employment contracts, with the government paying a share of the corresponding unemployment insurance entitlement and with reciprocal job-protection obligations (Ministério do Trabalho e Emprego, 2020). Short-time work schemes of this kind sustain headcount employment during transitory shocks but can impose reallocation costs when the shock is persistent (Giupponi and Landais, 2023).

What matters for the analysis below is that the two schemes were segmented and simultaneous. One required a formal contract; the other essentially required not having one. Both were large, and together they span the 2020Q2–2021Q3 window in which we find the education gradient reversing. Informality falls steeply with schooling: 40% of employed workers in our sample are informal, and informality is far more common among non-graduates (Ulyssea, 2020; Gerard and Gonzaga, 2021). A policy split along formality is therefore, mechanically, a policy split along education.

3. Data

3.1. PNAD Contínua and the matched panel

We use Brazil's *Continuous National Household Sample Survey* (*PNAD Contínua*), run by *IBGE*, which interviews roughly 211,000 households per quarter across about 16,000 census tracts. The survey uses a rotating panel: a sampled household is scheduled for five consecutive quarterly interviews, so consecutive quarterly samples overlap by four fifths.

PNAD Contínua does not publish an individual identifier, so respondents must be linked across interviews. We use the stage-3 identification of *Data Zoom*, which builds on Ribas and Soares (2008) and proceeds in three passes.[2] It first links respondents on household, sex and full date of birth; it then donates birth dates across interviews so that a respondent whose recorded date is missing or inconsistent in one interview is still matched on household position; and it finally resolves fragmented interview sequences by treating candidate links as a graph and taking connected components. The third pass is what recovers respondents whose recorded birth date drifts between interviews, and it is the reason the panel used here is longer and less selected than the household plus birth-date matching of the previous vintage of this paper.

Applied to the 52 quarterly surveys from 2012Q1 to 2024Q4, the algorithm links 3,024,050 individuals in 1,667,341 households across 39,270 primary sampling units. Surveys published after 2024Q4 enter the build too, but only as destinations: they complete the rotation groups still in the field and supply the $t+1$ state for workers whose origin quarter is 2024Q4. No quarter after 2024Q4 contributes an origin. Because the whole panel is rebuilt from the published microdata, the sampling unit, the stratum and the survey's own interview counter are carried through as ordinary columns. That matters twice over: it is what makes the design-based clustering of Section 4.3 possible, and it is what lets Section 6 distinguish a scheduled panel exit from attrition without having to infer a household's position in its rotation.

3.2. Employment exit

Our outcome is an employment exit: a worker employed at the interview date in quarter t who is no longer employed in $t+1$, that is, who is either unemployed or out of the labour force. We deliberately avoid the label "job loss". A transition out of employment is not necessarily a layoff: it can reflect retirement, a return to study, caring responsibilities, illness or discouragement, all of which were live margins during the pandemic. The term "employment exit" states what the data measure and nothing more.

Where a worker goes matters as much as whether they leave, both for mechanism and for policy, so we report the two destinations separately throughout. A worker who moves into measured unemployment is still

[2] *Data Zoom* is a data initiative of the Department of Economics at *PUC-Rio* that distributes cleaned Brazilian microdata through open-access *Stata* and *R* packages; we use the `advanced_3` panel of its *R* implementation.

searching; one who leaves the labour force is not. We therefore define

$$Y^{EU}_{i,t+1} = \mathbf{1}(E_t \to U_{t+1}), \qquad Y^{EN}_{i,t+1} = \mathbf{1}(E_t \to N_{t+1}). \tag{1}$$

The two indicators in (1) sum to the aggregate employment exit by construction, and each gets its own panel in Table 2. The split turns out to matter: the mid-pandemic reversal is 0.017 on the non-participation margin and 0.000 on the unemployment margin, so it is entirely a movement out of the labour force. Unconditionally, 2.9% of employed workers move into unemployment in the following quarter and 6.5% out of the labour force, so non-participation is the larger of the two margins. This matches Maurizio et al. (2023), who find that most Latin American informal workers who stopped working left the labour force rather than entering measured unemployment. We also report, as a complementary margin, an indicator for being informally employed in $t + 1$ (Table 8); note that this is a destination state rather than a transition, so it includes workers who were already informal in t and stayed.

Education enters as a binary indicator for completed tertiary education, which we call a college degree; 20% of employed workers hold one. Formality follows the standard *PNAD Contínua* classification, which varies by employment category: private employees with a signed work card and statutory public servants are formal, as are self-employed workers who contribute to social security and employers with a registered firm; their counterparts are informal. Appendix A gives the full variable dictionary.

3.3. Sample

The estimation sample is all workers aged 14 or over who are employed at t and matched to $t + 1$: 8,002,128 person-quarters, out of 11,491,554 employed person-quarters in total. The unmatched remainder is not discarded: it is retained in a companion file and is what Section 6 uses to model selection into the estimation sample directly. Table 1 reports survey-weighted descriptive statistics overall and by education group. The unconditional employment-exit rate is 9.4% per quarter, 10.7% among workers without a college degree and 4.0% among graduates. It is also much higher in informal than in formal employment (16.1% versus 4.9%), which is the first indication that formality, not schooling as such, is the operative margin.

Table 1: Descriptive statistics

	All workers		No college degree		College degree	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
Employment exit in $t+1$	0.094	0.292	0.107	0.309	0.040	0.197
Informally employed in $t+1$	0.328	0.469	0.364	0.481	0.180	0.385
College degree	0.197	0.398	0.000	0.000	1.000	0.000
Woman	0.427	0.495	0.396	0.489	0.552	0.497
Non-White	0.536	0.499	0.581	0.493	0.350	0.477
Urban area	0.876	0.330	0.853	0.354	0.970	0.170
Age	39.1	13.1	38.7	13.5	40.4	11.4
Formal employment	0.603	0.489	0.555	0.497	0.799	0.401
Informal employment	0.397	0.489	0.445	0.497	0.201	0.401
Formal private employee	0.393	0.488	0.401	0.490	0.361	0.480
Informal private employee	0.188	0.391	0.217	0.412	0.070	0.255
Formal public sector	0.099	0.299	0.055	0.228	0.280	0.449
Formal self-employed	0.076	0.265	0.073	0.260	0.088	0.284
Informal self-employed	0.174	0.379	0.198	0.398	0.074	0.262
Employer	0.044	0.205	0.036	0.185	0.078	0.268
Signed work card	0.407	0.491	0.410	0.492	0.394	0.489
Social security contributor	0.150	0.357	0.133	0.340	0.220	0.414
Temporary job	0.072	0.258	0.074	0.261	0.063	0.243
Usual weekly hours	39.3	12.1	39.4	12.4	38.7	10.4
Monthly labour income (R$)	2,189	3,627	1,506	1,965	4,973	6,433
Person-quarters	8,002,128		6,671,659		1,330,469	
Individuals	3,024,050		2,598,158		495,776	
Primary sampling units	39,270		39,245		34,998	

Notes: *PNAD Contínua* matched quarterly panel, 2012Q1–2024Q4. The unit of observation is a person-quarter for a worker employed at the interview date in quarter t who is matched to quarter $t+1$. Employment exit equals one when the worker is unemployed or out of the labour force in $t+1$. College degree indicates completed tertiary education. All moments are survey weighted using *PNAD Contínua* person weights.

4. Empirical strategy

4.1. Specification

We estimate a linear probability model in event-study form. For a worker i employed in quarter t,

$$Y_{i,t+1} \;=\; \alpha_t \;+\; \delta\,\text{College}_i \;+\; \sum_{q \neq 2019\text{Q4}} \beta_q\,(\text{College}_i \times \mathbf{1}\{t=q\}) \;+\; X'_{it}\gamma \;+\; \lambda_{s(it)} \;+\; \varepsilon_{i,t+1}, \tag{2}$$

where $Y_{i,t+1}$ indicates an employment exit, α_t are year-quarter fixed effects and $\lambda_{s(it)}$ collects fixed effects for state, urban status, gender, race, job tenure, occupation and sector. Occupation is coded by *IBGE*'s *Classificação de Ocupações para Pesquisas Domiciliares* (COD), the four-digit classification built for household surveys; we use its ten major groups, given by the leading digit.[3] Sector aggregates the survey's own activity grouping into five broad sectors: agriculture, industry, construction, trade and services. In both classifications a missing value is kept as an explicit "not reported" category rather than dropped, so that no observations are lost when they enter as fixed effects. The controls X_{it} are age, age squared, usual weekly hours, log labour income, formality, temporary-contract status and social security contribution. All estimates use *PNAD Contínua* person weights. The base quarter is 2019Q4, the last quarter whose $t+1$ outcome is measured before the shock, so every β_q is a deviation from the immediate pre-pandemic gap.

4.2. Adjusted margins

The education coefficients in equation (2) are conditional contrasts, not adjusted group means: $\widehat{\delta}$ is the college / non-college difference in the reference quarter and $\widehat{\delta}+\widehat{\beta}_q$ the difference in quarter q. Neither delivers the adjusted exit probability of either education group, and neither should be plotted as one. We therefore report survey-weighted average predictive margins for each education-quarter cell, together with their college / non-college differences:

$$\widehat{m}_{gq} \;=\; \frac{\sum_{i:t=q} w_i\,\widehat{\mathbb{E}}[Y_{i,t+1} \mid \text{College}_i = g,\, X_{it}]}{\sum_{i:t=q} w_i}, \qquad \widehat{\Delta}_q \;=\; \widehat{m}_{1q} - \widehat{m}_{0q}. \tag{3}$$

Equation (3) has a closed form, for two algebraic reasons. Because education enters only through its own indicator and its quarter interactions, the individual-level difference between the two counterfactuals is constant within a quarter, so $\widehat{\Delta}_q = \widehat{\delta} + \widehat{\beta}_q$ exactly, with variance read directly off the coefficient covariance matrix. The margins therefore add nothing to the contrasts, which the coefficients already deliver; what they add are the two levels. And because quarter is among the absorbed fixed effects, weighted residuals sum to zero within each quarter, so the two margins are $\widehat{m}_{gq} = \bar{y}_q + (g - p_q)\widehat{\Delta}_q$, where $\bar{y}_q$ is the survey-weighted exit rate and p_q the survey-weighted college share in quarter q. We verify numerically that this reproduces a general average-predictions routine to machine precision; the check is part of the replication package (Appendix A).

[3] See *IBGE, Estrutura da Classificação de Ocupações para Pesquisas Domiciliares* (COD).

4.3. Inference

Observations repeat individuals, are nested in households and *PNAD Contínua* primary sampling units, and share aggregate shocks within a quarter. Conventional standard errors treat those observations as independent, and are therefore not merely small but too small: at 8,002,128 records the understatement is large enough that almost any contrast would clear conventional thresholds. Our default variance estimator is therefore two-way clustered by primary sampling unit and by year-quarter (Cameron et al., 2011). The choice follows the design rather than the data, and the two dimensions answer different concerns. Abadie et al. (2023) show that clustering is called for when the sampling is itself clustered, which is exactly how *PNAD Contínua* draws its sample from a population of primary sampling units. Nothing is sampled along the time dimension, since we observe every quarter in the period; clustering on year-quarter instead addresses the aggregate shock common to everyone interviewed in a given quarter, which leaves the education gap correlated within quarters. Table B.1 shows the alternatives: by PSU, by household, by individual, by individual and year-quarter, and by year-quarter alone.

Because we report 52 quarterly contrasts, the probability that at least one pointwise 95% interval excludes zero by chance alone is far above 5%. We therefore also construct sup-t simultaneous bands by multiplier bootstrap over the joint distribution of the $\widehat{\Delta}_q$ (Montiel Olea and Plagborg-Møller, 2019), drawing 10,000 replications. The simultaneous critical value is 2.71 against a pointwise 1.96, and Figure 2 plots both bands.

For the two pandemic windows, where the number of effective time shocks rather than the number of records determines uncertainty, we additionally report wild cluster bootstrap p-values clustered by year-quarter (52 clusters), using Rademacher weights with the null imposed and 9,999 replications (Cameron et al., 2008; Roodman et al., 2019).

4.4. Decomposition

To ask whether any movement in the gradient reflects where the two groups work or what happens to them where they work, we decompose the unconditional education gap in exit rates into a composition and a within-cell term. The decomposition is the standardisation of Kitagawa (1955), of which Oaxaca (1973) and Blinder (1973) are the regression analogue; Fortin et al. (2011) survey the family of methods and the conditions under which the two components can be read separately. We write that gap Δ_q^{raw}, to distinguish it from the adjusted gap $\widehat{\Delta}_q$ of equation (3). Let k index cells defined by formality, sector and occupation, let s_{gkq} be the survey-weighted share of education group g in cell k in quarter q, and let m_{gkq} be that group's exit rate in the cell. Then

$$\Delta_q^{\text{raw}} = \underbrace{\sum_k (s_{Ckq} - s_{Nkq})\, m_{Nkq}}_{\text{composition}} + \underbrace{\sum_k s_{Ckq}\,(m_{Ckq} - m_{Nkq})}_{\text{within-cell}}, \tag{4}$$

with C and N denoting college and non-college. Like any two-term decomposition of this family, (4) fixes a reference: the composition term holds the non-college exit rates fixed and the within-cell term holds the college allocation fixed. We report the whole exercise a second time on the finer partition that also splits cells by labour-market position, in the lower panel of Table 5; the conclusions are unchanged. Standard errors and 95% percentile intervals come from a cluster bootstrap that draws exponential multipliers at the PSU-by-quarter level, 500 replications.

5. Results

5.1. The education gradient over time

Figure 1 plots the adjusted employment-exit probability for the two education groups quarter by quarter, Figure 2 plots the difference with pointwise and simultaneous bands, and Table 2 summarises the same objects by period, together with each group's change relative to its own pre-pandemic level.

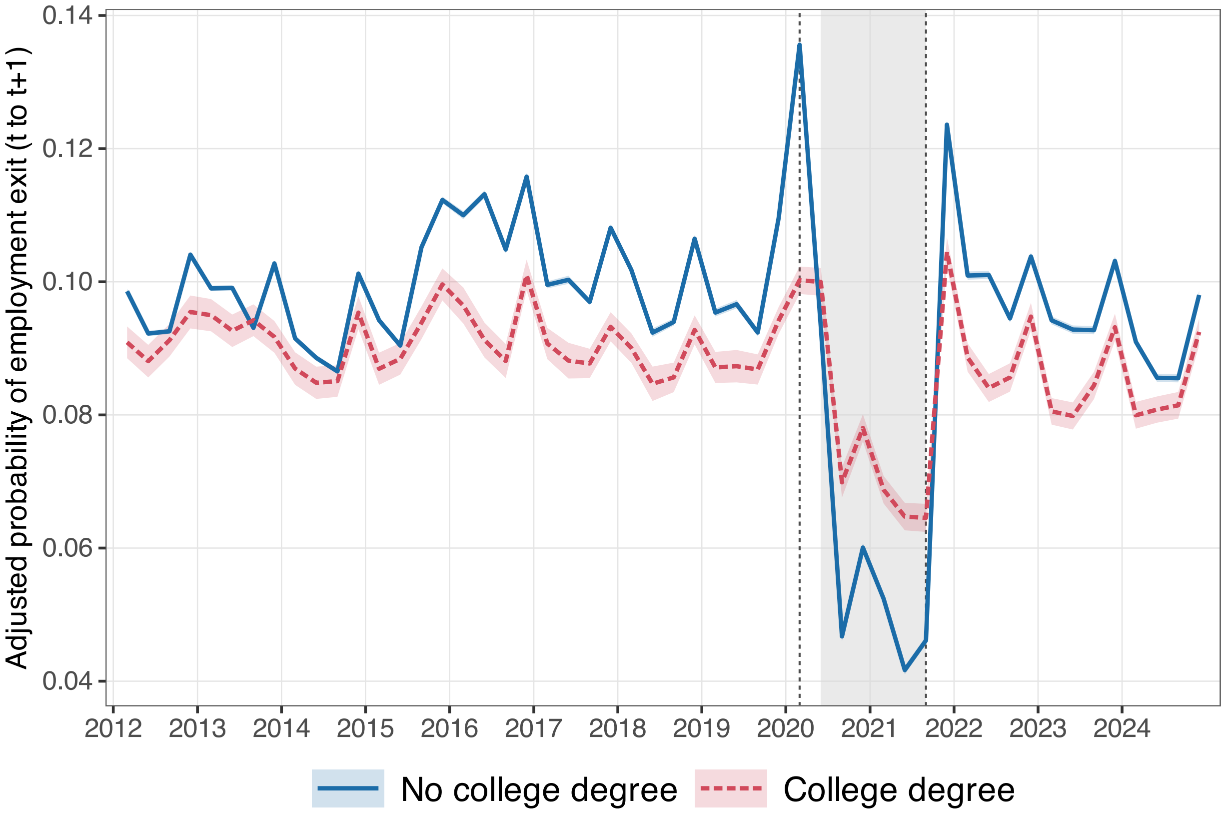

Figure 1: Adjusted probability of employment exit by education, 2012Q1–2024Q4. Survey-weighted average predictive margins from equation (2), with 95% pointwise bands. The shaded band marks 2020Q2–2021Q3; the dashed lines mark 2020Q1 and 2021Q3.

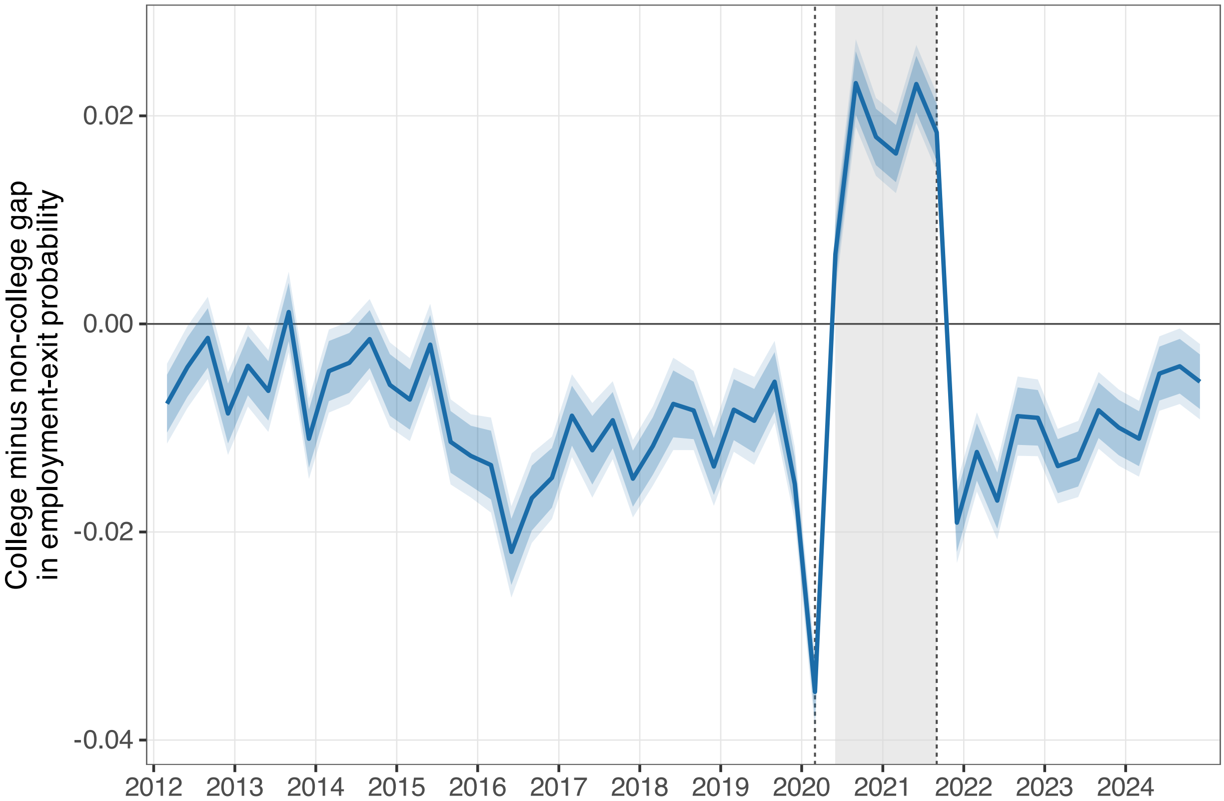

Figure 2: College minus non-college difference in the adjusted probability of employment exit. The dark band is the pointwise 95% interval; the light band is the sup-t simultaneous band over all 52 quarterly contrasts (critical value 2.71).

Table 2: Adjusted employment-exit probabilities and the college gap, by destination and period

	Adjusted level		College − no college			
	No college	College	Difference	95% CI	% change vs. pre-pandemic	
Period	(1)	(2)	(3)	(4)	No college	College
Panel A. Employment exit						
Pre-pandemic	0.100	0.091	-0.009***	[-0.011, -0.006]	–	–
	(0.0002)	(0.0011)	(0.0013)			
Onset	0.136	0.100	-0.035***	[-0.038, -0.033]	36.1	10.4
	(0.0003)	(0.0011)	(0.0014)			
Mid-pandemic	0.056	0.074	0.018***	[0.015, 0.020]	-43.3	-18.3
	(0.0003)	(0.0010)	(0.0013)			
Post-pandemic	0.097	0.087	-0.010***	[-0.013, -0.008]	-2.5	-4.4
	(0.0003)	(0.0010)	(0.0013)			
Panel B. Exit into unemployment						
Pre-pandemic	0.032	0.028	-0.003***	[-0.005, -0.002]	–	–
	(0.0001)	(0.0005)	(0.0007)			
Onset	0.041	0.028	-0.012***	[-0.014, -0.011]	28.3	-0.4
	(0.0002)	(0.0005)	(0.0007)			
Mid-pandemic	0.023	0.023	0.000	[-0.001, 0.002]	-27.6	-18.0
	(0.0002)	(0.0005)	(0.0006)			
Post-pandemic	0.026	0.025	-0.001	[-0.002, 0.000]	-16.6	-10.1
	(0.0001)	(0.0005)	(0.0006)			
Panel C. Exit into non-participation						
Pre-pandemic	0.068	0.062	-0.006***	[-0.008, -0.004]	–	–
	(0.0002)	(0.0009)	(0.0010)			
Onset	0.095	0.072	-0.023***	[-0.025, -0.021]	39.8	15.3
	(0.0003)	(0.0009)	(0.0011)			
Mid-pandemic	0.034	0.051	0.017***	[0.015, 0.019]	-50.7	-18.5
	(0.0003)	(0.0008)	(0.0011)			
Post-pandemic	0.071	0.061	-0.009***	[-0.011, -0.007]	4.1	-1.8
	(0.0002)	(0.0008)	(0.0010)			

Notes: Entries in columns (1)–(3) are survey-weighted average predictive margins from the event-study specification of equation (2), which includes year-quarter, state, urban, gender, race, tenure, occupation and sector fixed effects and controls for age, age squared, usual weekly hours, log labour income, formality, temporary status and social security contribution. Margins are computed by holding the covariate distribution fixed and setting the college indicator to zero and to one in turn; they are not regression intercepts. Panel A is any exit from employment; Panels B and C split it by destination, so B and C sum to A by construction. Standard errors in parentheses are two-way clustered by primary sampling unit and year-quarter. Stars on the difference in column (3) denote $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; the adjusted levels are probabilities and carry none. The last two columns report the percentage change in each group's adjusted level relative to its own pre-pandemic average.

The series falls into three phases. Through 2012–2019 the gradient is stable and runs the expected way:

the adjusted gap averages -0.009 (s.e. 0.001), so graduates are about 0.9 percentage points less likely to leave employment than otherwise comparable non-graduates. At the onset, in 2020Q1, the gradient widens sharply to -0.035: the first wave of exits fell disproportionately on workers without a degree, whose adjusted exit level rose 36% above its pre-pandemic average, against a rise of 10% for graduates. This contrast, however, is identified from a single quarter. Its asymptotic standard error is small under every clustering scheme in Table B.1, including clustering on year-quarter alone, but that is exactly the configuration in which the cluster-robust variance is known to be badly downward biased: the deviation is picked up in one cluster out of 52 (Cameron et al., 2008; Roodman et al., 2019). Imposing the null and resampling signs across quarters, the wild cluster bootstrap fails to reject at conventional levels (Table 3). Put plainly, a movement of this size in the education gap in a single quarter is not unusual once quarterly shocks to the gap are taken seriously. We report the onset estimate as a description of what happened in 2020Q1 and build nothing on it.

From 2020Q2 the gradient inverts. Over the six quarters to 2021Q3 the adjusted gap averages 0.018 (s.e. 0.001, 95% CI [0.015, 0.020]), peaking at 0.023 in 2020Q3. The estimated gap is positive in 7 of the 52 quarters: the six inside the window, plus an isolated 2013Q3. The simultaneous band excludes zero in 6 of those 7, all of them inside the window, so the reversal is not an artefact of testing 52 contrasts one at a time. Mechanically the inversion comes from both sides: the adjusted exit level of non-graduates falls 43% below its pre-pandemic average while that of graduates falls only 18%. From 2021Q4 the pre-pandemic configuration returns, and by the end of the sample the adjusted gap is -0.010.

Table 3 shows how the two pandemic contrasts move as controls and fixed effects are added, and reports wild cluster bootstrap p-values clustered by year-quarter for the preferred specification. The ladder is informative in two ways. Substantively, the reversal appears only in columns (5) and (6): with quarter, state and demographic controls alone the mid-pandemic gap is still negative, and it turns positive once time-varying job characteristics and occupation and sector fixed effects are added. This is the same message the decomposition delivers in Section 5.3. Statistically, the mid-pandemic contrast is unambiguous under the year-quarter bootstrap while the onset contrast is not, for the reason just given: one treated quarter cannot be separated from a quarterly shock.

Table 3: The college gap in employment exits across specifications

	(1)	(2)	(3)	(4)	(5)	(6)
College, other quarters (δ)	-0.069***	-0.069***	-0.060***	-0.053***	-0.023***	-0.010***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
College × onset (ω_1)	-0.021***	-0.021***	-0.023***	-0.025***	-0.025***	-0.026
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Wild cluster bootstrap p	–	–	–	–	–	0.177
College × mid-pandemic (ω_2)	0.029***	0.029***	0.029***	0.027***	0.027***	0.027***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Wild cluster bootstrap p	–	–	–	–	–	<0.0001
Implied college gap						
Onset ($\delta + \omega_1$)	-0.090***	-0.090***	-0.083***	-0.077***	-0.048***	-0.035***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Mid-pandemic ($\delta + \omega_2$)	-0.040***	-0.040***	-0.030***	-0.026***	0.005**	0.018***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Year-quarter FE	–	Yes	Yes	Yes	Yes	Yes
State and urban FE	–	–	Yes	Yes	Yes	Yes
Gender and race FE	–	–	–	Yes	Yes	Yes
Tenure FE	–	–	–	–	Yes	Yes
Occupation and sector FE	–	–	–	–	–	Yes
Time-varying job controls	–	–	–	–	Yes	Yes
Observations	8,002,128	8,002,128	8,002,128	8,002,128	8,002,128	8,002,128
R^2	0.010	0.011	0.020	0.045	0.091	0.093

Notes: Survey-weighted linear probability models. The dependent variable is an indicator for exiting employment between t and $t+1$. δ is the college gap outside the two pandemic windows; ω_1 and ω_2 shift it at the onset and during the mid-pandemic window. Standard errors in parentheses are two-way clustered by primary sampling unit and year-quarter. For the two pandemic deviations we also report, for the preferred specification in column (6), wild cluster bootstrap p-values clustered by year-quarter (52 clusters), using Rademacher weights with the null imposed and 9,999 replications; they test ω_1 and ω_2, the shifts in the gap, which is where the number of quarterly shocks rather than the number of records determines uncertainty. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; they are based on the wild cluster bootstrap p-value wherever one is reported, and on the asymptotic two-way clustered standard error otherwise. The onset deviation ω_1 therefore carries no stars in column (6): identified from a single year-quarter, it cannot be told apart from an ordinary quarterly shock. The implied gaps keep their asymptotic stars because they are dominated by δ, which is identified across all 52 quarters. Periods are pre-pandemic 2012Q1–2019Q4, onset 2020Q1, mid-pandemic 2020Q2–2021Q3 and post-pandemic 2021Q4–2024Q4.

5.2. Where the reversal happens

Figure 3 re-estimates the model separately on formal and on informal employment.

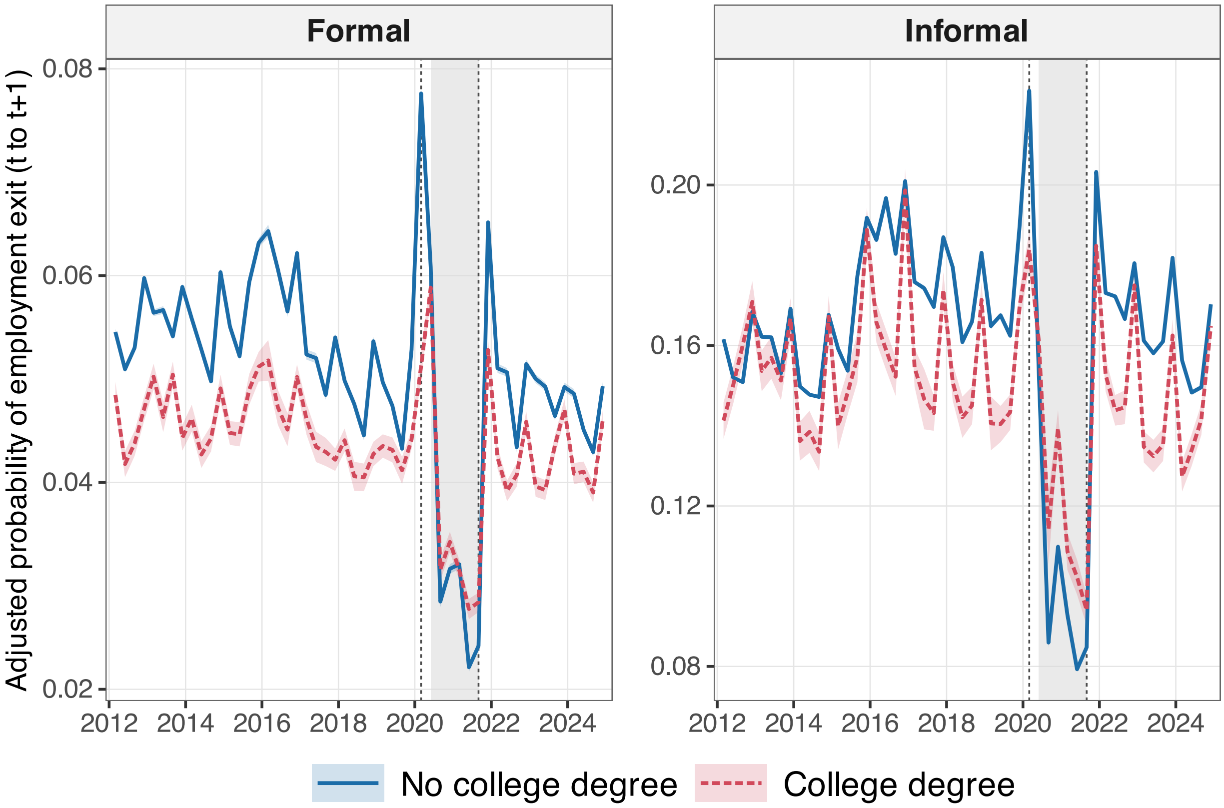

Figure 3: Adjusted probability of employment exit by education, formal and informal employment estimated separately. Note the different vertical scales.

Exit risk is far higher in informal employment throughout, and the reversal is an informal-sector phenomenon. Among informal workers the adjusted gap moves from -0.014 before the pandemic to 0.019 (s.e. 0.002) in 2020Q2–2021Q3. Among formal workers it moves from -0.009 to 0.002. Splitting each side into detailed labour-market positions narrows the locus further. The reversal is driven by informal private employees, workers in a dependent employment relationship without a signed work card, whose adjusted gap moves from -0.015 to 0.029. It is absent among the informal self-employed (-0.001), among formal private employees (0.000) and among formal public servants (-0.003); only the formal self-employed show a smaller echo of it (0.008). Figure 4 shows the four groups separately, and Table 4 reports all segments.

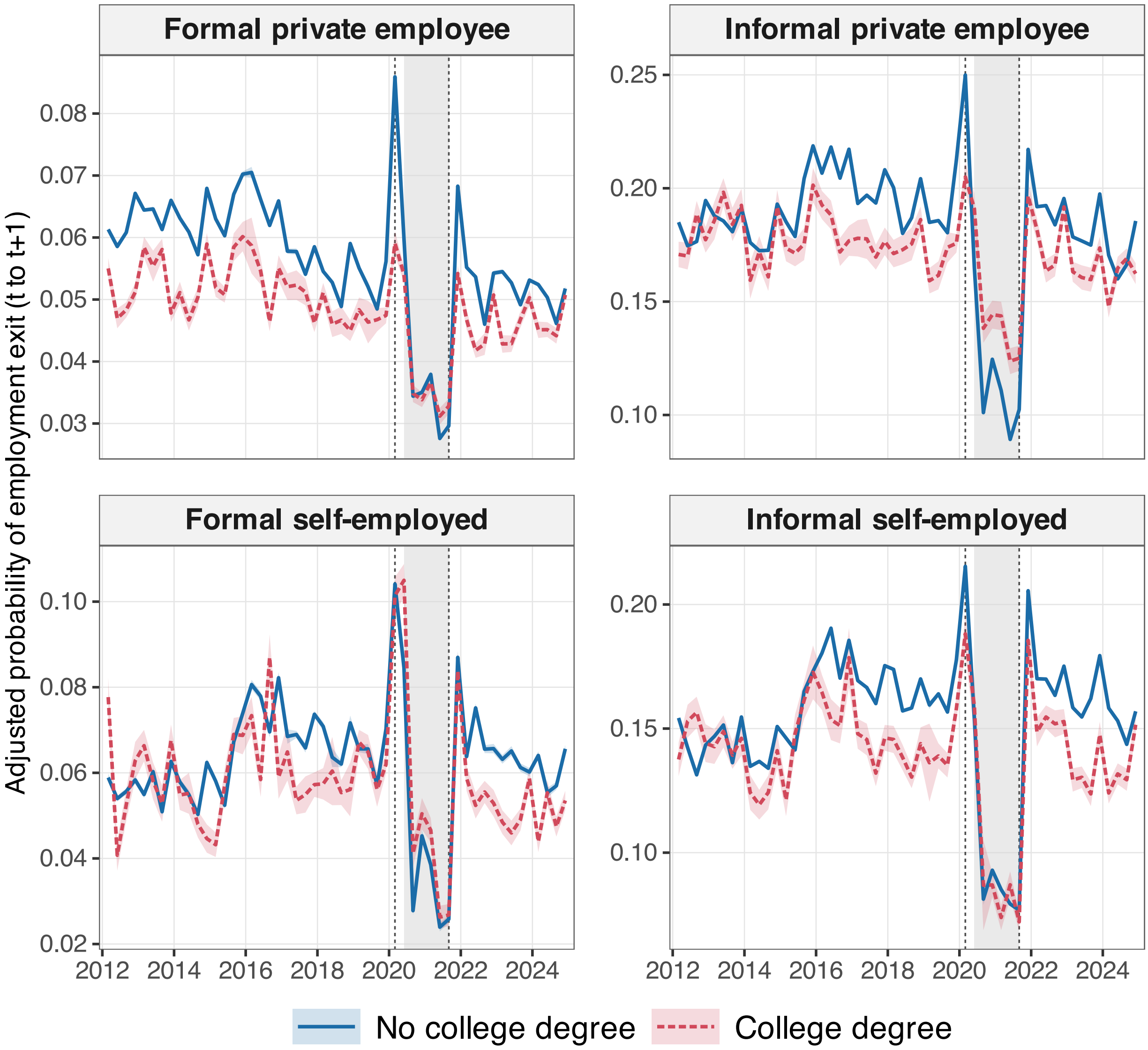

Figure 4: Adjusted probability of employment exit by education and labour market position. Each panel is estimated on its own subsample; public-sector employees are reported separately in Table 4. Note the different vertical scales.

Table 4: Adjusted college gap in employment exits, by labour market segment

Segment	Pre-pandemic	Onset	Mid-pandemic	Post-pandemic	Obs.
Formality					
Formal	-0.009***	-0.026***	0.002***	-0.006***	4,363,453
	(0.001)	(0.001)	(0.001)	(0.001)	
Informal	-0.014***	-0.040***	0.019***	-0.019***	3,638,675
	(0.002)	(0.003)	(0.002)	(0.002)	
Gender					
Men	-0.004***	-0.029***	0.015***	-0.005***	4,656,560
	(0.001)	(0.001)	(0.001)	(0.001)	
Women	-0.013***	-0.043***	0.025***	-0.016***	3,345,568
	(0.002)	(0.002)	(0.002)	(0.002)	
Position					
Formal private employee	-0.009***	-0.027***	0.000	-0.006***	2,679,329
	(0.001)	(0.001)	(0.001)	(0.001)	
Formal public sector	-0.004***	-0.007***	-0.003***	-0.007***	818,008
	(0.001)	(0.001)	(0.001)	(0.001)	
Formal self-employed	-0.005***	-0.003	0.008***	-0.011***	624,666
	(0.002)	(0.003)	(0.002)	(0.001)	
Informal private employee	-0.015***	-0.045***	0.029***	-0.015***	1,674,121
	(0.002)	(0.004)	(0.003)	(0.002)	
Informal self-employed	-0.014***	-0.027***	-0.001	-0.022***	1,621,692
	(0.002)	(0.004)	(0.003)	(0.002)	
Race					
Non-White	-0.013***	-0.032***	0.022***	-0.013***	4,582,254
	(0.001)	(0.002)	(0.002)	(0.001)	
White	-0.009***	-0.034***	0.010***	-0.009***	3,419,874
	(0.001)	(0.001)	(0.001)	(0.001)	

Notes: Each cell is the survey-weighted average predictive difference in the probability of exiting employment between workers with and without a college degree, estimated on the indicated subsample with the full specification of equation (2). Fixed effects and controls that are collinear within a segment are dropped. Standard errors in parentheses are two-way clustered by primary sampling unit and year-quarter. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Periods are pre-pandemic 2012Q1–2019Q4, onset 2020Q1, mid-pandemic 2020Q2–2021Q3 and post-pandemic 2021Q4–2024Q4.

That the movement is concentrated among informal employees rather than the informal self-employed is informative. Dismissal risk is the margin that moves, and it moves for the unprotected wage workers whose separations are employer decisions, not for own-account workers whose exits are largely their own. It also means the pattern cannot be read as a pure labour-supply response to the transfer, which would have been at least as visible among the self-employed.

5.3. Composition or within-cell risk?

The decomposition in equation (4) separates the two explanations. Figure 5 plots the components quarter by quarter and Table 5 reports period averages.

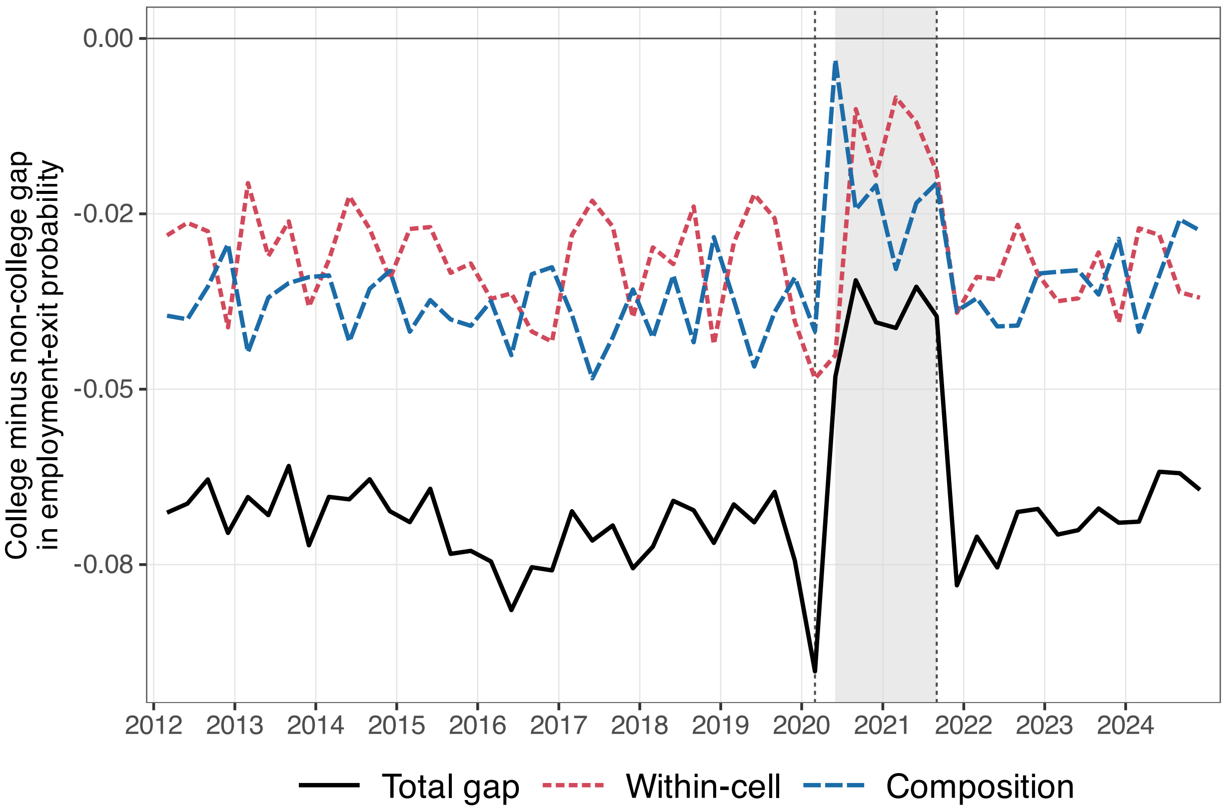

Figure 5: Composition and within-cell components of the unconditional education gap in employment exits. Cells are defined by formality × sector × occupation.

Table 5: Composition versus within-cell decomposition of the education gap

Period	Total gap Δ_q^{raw}	Composition component	Within-cell component	Within-cell share (%)
Cells: formality × sector × occupation (100 cells)				
Pre-pandemic	-0.069***	-0.038***	-0.031***	44.9
		[-0.039, -0.037]	[-0.033, -0.030]	
Onset	-0.090***	-0.042***	-0.048***	53.7
		[-0.054, -0.030]	[-0.062, -0.036]	
Mid-pandemic	-0.040***	-0.021***	-0.019***	47.5
		[-0.025, -0.017]	[-0.023, -0.015]	
Post-pandemic	-0.069***	-0.035***	-0.034***	49.5
		[-0.037, -0.032]	[-0.036, -0.031]	
Cells: formality × position × sector × occupation (382 cells)				
Pre-pandemic	-0.069***	-0.042***	-0.028***	39.8
		[-0.043, -0.040]	[-0.029, -0.026]	
Onset	-0.090***	-0.052***	-0.038***	42.2
		[-0.062, -0.042]	[-0.049, -0.028]	
Mid-pandemic	-0.040***	-0.024***	-0.016***	40.6
		[-0.027, -0.020]	[-0.020, -0.012]	
Post-pandemic	-0.069***	-0.037***	-0.031***	45.3
		[-0.040, -0.035]	[-0.033, -0.029]	

Notes: The total gap is the survey-weighted difference in the raw employment-exit rate between workers with and without a college degree. The composition component holds each cell's non-college exit rate fixed and varies only the two groups' allocation across cells; the within-cell component holds the college allocation fixed and varies only the exit rates inside cells. Period figures are averages of the quarterly components weighted by the quarter's share of the total survey weight. Brackets report 95% percentile intervals from 500 bootstrap replications that draw exponential multipliers at the primary-sampling-unit-by-quarter level. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ on the bootstrap standard error of the entry. Periods are pre-pandemic 2012Q1–2019Q4, onset 2020Q1, mid-pandemic 2020Q2–2021Q3 and post-pandemic 2021Q4–2024Q4.

Neither component dominates, and that holds under both cell definitions. Composition accounts for 55% of the unconditional pre-pandemic gap (-0.038 out of -0.069) and within-cell risk for the remaining 45% (-0.031), and those shares are essentially unchanged in every other period. Graduates are safer partly because they hold formal jobs in more protected occupations and sectors, and partly because they are safer inside those jobs.

What happens to the two components during the pandemic is more informative: they compress together during 2020Q2–2021Q3, composition by 45% and within-cell risk by 39%. The unconditional gap therefore halves, from -0.069 to -0.040, without changing sign, and the split between the two components is the same before, during and after. The mid-pandemic narrowing is not a reallocation story: graduates did not move into riskier cells.

This pins down the reversal of Figure 2. It is not visible at the level of formality, sector and occupation, where the education advantage merely shrinks by about half. It emerges only once the richer conditioning set of equation (2) is applied, that is, once two workers are matched not only on cell but also on hours, labour income, tenure, contract type, social security status and location. Inside a given formality-sector-occupation cell, graduates still hold the better-paid, longer-tenured, contractually protected jobs. Strip that residual advantage away as well and, for six quarters, their edge in staying employed does not merely disappear: it turns negative.

5.4. Gender and race

Table 4 and Figure 6 report the same adjusted margins by gender and by race. The pre-pandemic gradient is not uniform: it is markedly steeper for women (-0.013) than for men (-0.004), and steeper for non-white workers (-0.013) than for white workers (-0.009). The mid-pandemic reversal nonetheless appears in every group, at 0.015 for men and 0.025 for women, and 0.010 for white and 0.022 for non-white workers. The aggregate pattern is therefore not the product of a single demographic group. Its magnitude, however, tracks informality: it is largest among women and among non-white workers, the two groups whose employment is most concentrated in the informal segment, which is what the reading in Section 5.2 implies.

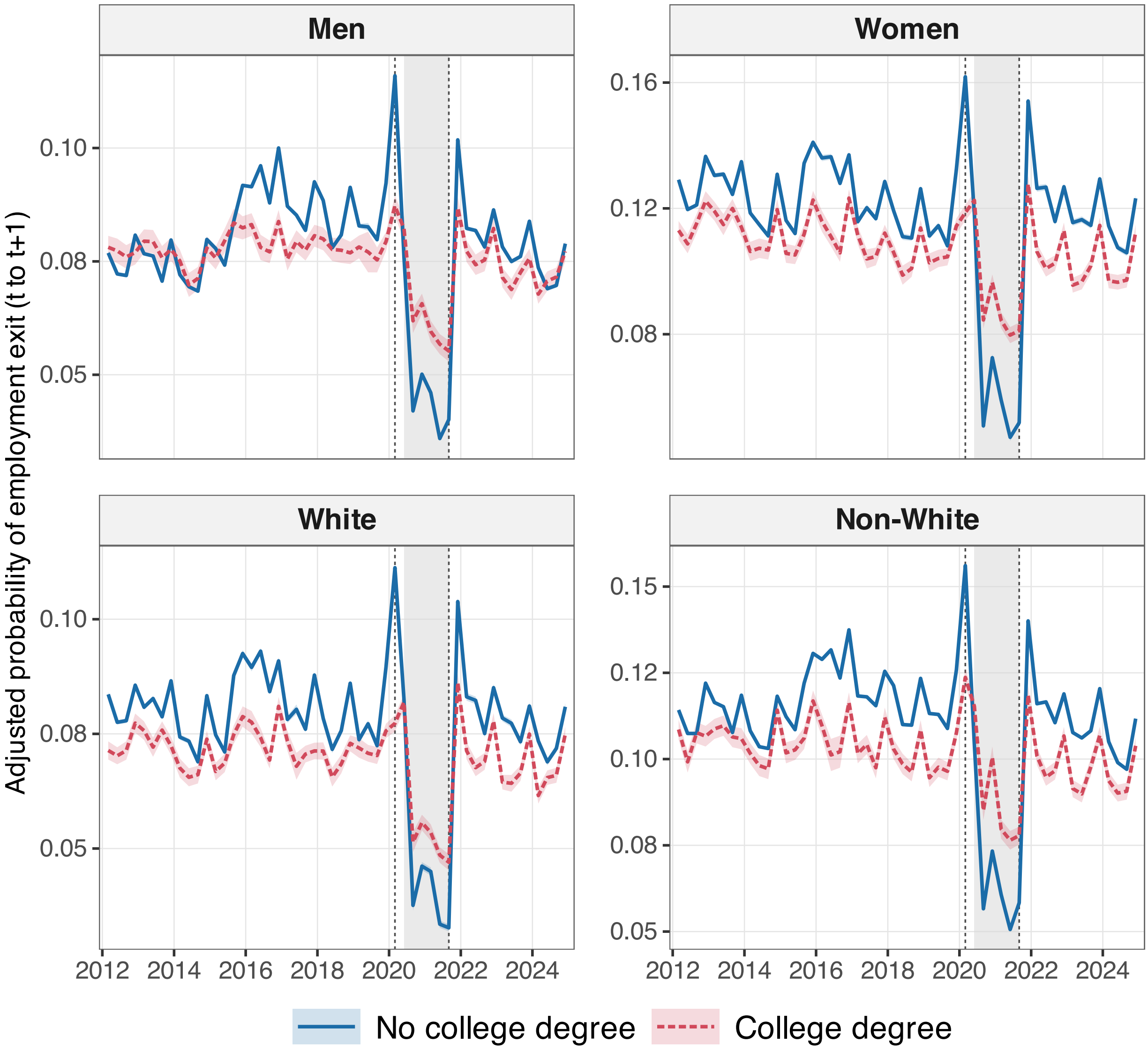

Figure 6: Adjusted probability of employment exit by education, within demographic groups. Each panel is estimated on its own subsample, with 95% pointwise bands. Note the different vertical scales.

6. Panel retention and selection

The outcome exists only for workers the matching algorithm finds again, so differential retention is a first-order concern: if graduates and non-graduates became differentially harder to re-interview in 2020–2021, part of the reversal could be an artefact.

The selection that generates the estimation sample is the match from t into $t+1$, and here it is observed directly. The build keeps every worker employed in quarter t, whether or not the algorithm finds them again, so the indicator we model is the very object at issue rather than a stand-in for it.

One restriction applies. A household is interviewed five consecutive quarters, so a worker observed in the fifth interview has no $t+1$ inside the panel: that is a scheduled exit, not attrition. The survey's own interview counter identifies those rows, and we drop them. Figure 7 plots the resulting match rate and Table 6 summarises it.

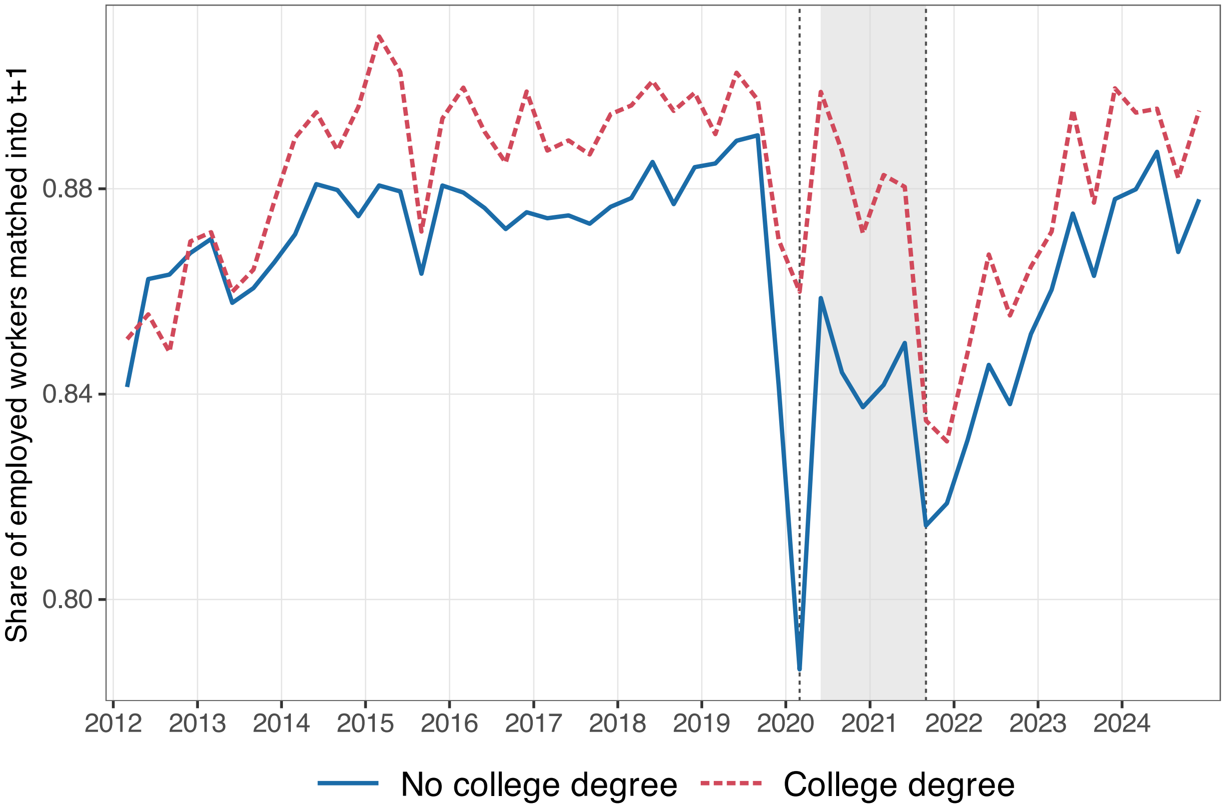

Figure 7: Share of employed workers matched into the following quarter, by education. Sample restricted to interviews 1–4, the interviews for which a further one is scheduled.

Graduates are consistently easier to re-find, and the pandemic widened the gap. The college / non-college difference in the match rate is 0.014 before the pandemic, jumps to 0.074 in 2020Q1 when fieldwork switched to telephone interviewing, and is still 0.035 through 2020Q2–2021Q3 before settling back. This is the pattern that should make a reader uneasy: the workers we lose track of are disproportionately non-graduates, and we lose more of them in the very quarters where the gradient moves. Panel B of Table 6 shows the same thing in levels, with the largest standardised differences on formality and labour income.

We do not dismiss the concern by appealing to its direction, because the direction is unfavourable. The workers who drop out of view carry the higher exit risk, and they are disproportionately non-graduates, so the exit rate we measure for non-graduates is too low and the mid-pandemic gap we estimate is, if anything, too positive. Differential retention pushes towards the reversal, not against it, and has to be bounded rather than signed away. We bound it twice: by re-weighting on the observables that drive retention, and by asking how large the unobserved differential would have to be to overturn the result.

Table 6: Panel retention and selection into the matched sample

Panel A. Share of employed workers matched into $t+1$

	By education			By formality		
Period	No college	College	Diff.	Formal	Informal	Diff.
Pre-pandemic	0.873	0.887	0.014	0.877	0.872	0.005
Onset	0.786	0.860	0.074	0.823	0.771	0.052
Mid-pandemic	0.841	0.876	0.035	0.856	0.838	0.018
Post-pandemic	0.860	0.876	0.016	0.866	0.860	0.006

Panel B. Matched vs. unmatched origins, all quarters

	Matched	Unmatched	Std. diff.
College degree	0.197	0.174	0.059
Woman	0.427	0.409	0.036
Non-White	0.536	0.586	-0.101
Urban area	0.876	0.899	-0.074
Age	39.067	35.725	0.262
Usual weekly hours	39.307	39.994	-0.058
Log labour income	6.908	6.937	-0.017
Formal employment	0.603	0.587	0.033
Signed work card	0.407	0.436	-0.058
Social security contributor	0.150	0.120	0.090
Temporary job	0.072	0.079	-0.026

Notes: The sample is every worker employed at the interview date in quarter t who is not in the fifth and final interview of their household's rotation, so that a further interview is scheduled. Retention equals one when the stage-3 algorithm links the worker into $t+1$. Panel B reports survey-weighted means and the standardised difference, $(\bar{x}_1 - \bar{x}_0)/\sqrt{(s_1^2 + s_0^2)/2}$. Periods are pre-pandemic 2012Q1–2019Q4, onset 2020Q1, mid-pandemic 2020Q2–2021Q3 and post-pandemic 2021Q4–2024Q4.

We then re-weight. A separate logit per quarter, on education, demographics, job characteristics and state, occupation and sector fixed effects, gives each worker a predicted probability of being matched forward; fitting it quarter by quarter lets every coefficient, not only the one on education, move over time. Multiplying the survey weight by the inverse of that probability, trimmed at 0.05, and re-estimating equation (2) leaves the mid-pandemic gap essentially unchanged: 0.0179 with the re-weighting, against 0.0177 without it on the same sample. The two agree to three decimals, which is why Table 7 reports four: re-weighting shifts each period's gap by less than half a thousandth, and column (3) gives the shift directly.

Table 7: The adjusted college gap under inverse-retention-probability weighting

Period	Survey weights (1)	Survey × inverse-retention weights (2)	Difference (2) − (1) (3)
Pre-pandemic	-0.0089***	-0.0089***	-0.0001
	(0.0013)	(0.0014)	
Onset	-0.0354***	-0.0353***	0.0000
	(0.0014)	(0.0014)	
Mid-pandemic	0.0177***	0.0179***	0.0003
	(0.0013)	(0.0014)	
Post-pandemic	-0.0104***	-0.0103***	0.0001
	(0.0013)	(0.0013)	

Notes: Both columns are estimated on the matched origins of the retention sample (8,002,128 person-quarters) with the full specification, so the two differ only in the weights. Column (2) multiplies the survey weight by the inverse of a predicted probability of being matched into $t+1$, trimmed at 0.05, from a separate logit per year-quarter on education, demographics, job characteristics and state, occupation and sector fixed effects. Column (3) reports the difference between the two point estimates. Entries are given to four decimals because re-weighting moves the gap by less than half a thousandth. Standard errors are two-way clustered by primary sampling unit and year-quarter. Periods are pre-pandemic 2012Q1–2019Q4, onset 2020Q1, mid-pandemic 2020Q2–2021Q3 and post-pandemic 2021Q4–2024Q4.

A breakdown calculation then bounds what unobserved attrition could do. The algorithm links 86.8% of the employed workers for whom a further interview is scheduled into $t+1$ (Table A.1), leaving the rest unobserved. Writing ρ for that match rate, the education gap among all such workers is $\rho\widehat{\Delta}^{\text{mid}} + (1-\rho)(\tilde{m}_C - \tilde{m}_N)$, where $\widehat{\Delta}^{\text{mid}}$ is the estimated mid-pandemic gap and $\tilde{m}_g$ is the unobserved exit rate of unmatched workers in group g. Setting that expression to zero requires an unmatched-worker exit-rate differential of $\tilde{m}_C - \tilde{m}_N = -0.1160$, opposite in sign to what is observed among matched workers and several times larger in magnitude. We regard that as implausible, though not impossible. The calculation applies a common ρ to both education groups; Panel A of Table 6 reports the modest differential in retention between them.

7. Robustness

Table 8 collects the remaining checks. Restricting to prime-age workers aged 25–54 removes retirement and schooling transitions from the outcome and leaves the pattern intact. Dropping the survey weights and moving the base quarter to 2019Q1 do the same. Replacing the outcome by an indicator for being informally employed in $t+1$ shows the complementary margin of adjustment. There the education gradient runs the usual way throughout and widens during the window, from -0.028 to -0.047: graduates who kept working became relatively less exposed to informal employment just as their advantage in staying employed at all disappeared. Table B.1 shows that the two pandemic contrasts are robust to the choice of variance estimator, including clustering on year-quarter alone, which is the most conservative option available given 52 clusters.

Table 8: Robustness of the adjusted college gap

Specification	Pre-pandemic	Onset	Mid-pandemic	Post-pandemic	Obs.
Baseline	-0.009***	-0.035***	0.018***	-0.010***	8,002,128
	(0.001)	(0.001)	(0.001)	(0.001)	
Prime age 25–54	-0.008***	-0.037***	0.009***	-0.012***	5,570,238
	(0.001)	(0.001)	(0.001)	(0.001)	
Unweighted	-0.011***	-0.026***	0.024***	-0.015***	8,002,128
	(0.001)	(0.001)	(0.001)	(0.001)	
Base quarter 2019Q1	-0.009***	-0.035***	0.018***	-0.010***	8,002,128
	(0.001)	(0.001)	(0.001)	(0.001)	
Outcome: informal employment in $t+1$	-0.028***	-0.009***	-0.047***	-0.020***	8,002,128
	(0.002)	(0.002)	(0.002)	(0.002)	

Notes: Each row reports survey-weighted average predictive differences in the outcome between workers with and without a college degree, from the full specification of equation (2). The last row replaces the outcome by an indicator for being informally employed in $t+1$ – a destination state, not a transition, so it includes workers who were already informal in t. Standard errors in parentheses are two-way clustered by primary sampling unit and year-quarter. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Periods are pre-pandemic 2012Q1–2019Q4, onset 2020Q1, mid-pandemic 2020Q2–2021Q3 and post-pandemic 2021Q4–2024Q4.

8. Discussion

The results say something specific. Schooling protects Brazilian workers from employment exits through two channels of comparable size: placement into formal jobs in more protected occupations and sectors, and a residual advantage inside those cells. Neither dominates, and their relative weights are stable over time. During 2020Q2–2021Q3 both compress by roughly the same proportion, halving the unconditional gap without reversing it. The reversal appears only when workers are also matched on the job characteristics that vary within cells, such as pay, hours, tenure and contract type, and it appears in informal employment, specifically among informal private employees.

We are correspondingly careful about the onset. Its point estimate is large, but a single quarter carries no information that survives clustering on time, so nothing in what follows rests on it. The timing of the reversal is consistent with, but does not identify, a role for the emergency transfer and employment-preservation programmes. *Auxílio Emergencial* started in April 2020 and ended in October 2021, bracketing the window almost exactly, and it reached the informal, lower-education workers whose adjusted exit risk fell most. The employment-preservation programme worked on the formal side, where we find much smaller movements. But the design here has no eligibility discontinuity and no pre-determined exposure measure, so we cannot separate the programmes from everything else that moved at the same time: sectoral shutdowns, differential remote-work feasibility (Dingel and Neiman, 2020), changed labour supply of graduates with children (Alon et al., 2020), or the composition of who kept searching. The United States experience with pandemic unemployment benefits illustrates how sharply such programmes can alter measured transitions even when they are not designed to (Ganong et al., 2020; Marinescu et al., 2021). Establishing the policy channel would require variation in eligibility or in pre-determined exposure, which we do not have. Claims that "policy design matters" go beyond what this evidence supports; what it supports is that the education gradient in job security is not a structural constant, and that it moved in the segment of the labour market where the segmented policy response was concentrated.

A limitation remains. The design is descriptive. We observe that the gradient reversed in informal wage employment over the window in which the emergency transfer operated, but we do not observe anyone who was ineligible for it. Nothing here rules out a coincident shock with the same timing and the same incidence.

The transferable lesson is about the architecture of emergency support rather than about Brazil. When protection is delivered through two channels that are segmented along the formal–informal margin, and when schooling predicts which channel a worker falls into, the aggregate relationship between education and job security can temporarily detach. Any country with a large informal sector and a bifurcated safety net has the same structure.

9. Conclusion

Using 8,002,128 matched person-quarters from *PNAD Contínua* over 2012Q1–2024Q4, we document that Brazil's education gradient in employment exits reversed for six quarters between 2020Q2 and 2021Q3 and then returned to its pre-pandemic configuration. The single-quarter widening at the onset, though large in point estimate, does not survive inference clustered on time and we do not build on it. The reversal is present in adjusted margins computed with design-based clustering and simultaneous bands, runs entirely through exits out of the labour force rather than into measured unemployment, is concentrated in informal wage employment, and survives weighting by the inverse probability of being matched into the following quarter. A composition / within-cell decomposition shows that the education advantage divides about evenly between where graduates work and how they fare inside those jobs, and that both halves compressed by a similar proportion, 45% and 39%, during the window in which Brazil's emergency transfer was operating. The reversal is therefore a statement about workers matched on job characteristics, not about reallocation across sectors. The episode is a reminder that the protective value of schooling runs through the structure of the labour market, and that this structure can shift fast enough to suspend the gradient for six quarters.

Declarations

CRediT author statement. **Francisco Cavalcanti:** Conceptualization, Methodology, Supervision, Writing – review and editing. **Fredie Didier:** Conceptualization, Data curation, Formal analysis, Software, Visualization, Writing – original draft. **Gustavo Gonzaga:** Conceptualization, Methodology, Validation, Writing – review and editing.

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI use. During the preparation of this work the authors used generative AI tools to assist with code refactoring and language editing. The authors reviewed and edited all output and take full responsibility for the content of the publication.

Data availability. *PNAD Contínua* microdata are public and distributed by *IBGE*. The rotating-panel links are built with the open-source *Data Zoom* package. All code needed to reproduce every table and figure, together with the variable dictionary, matching rules, package versions and random seeds, is available in the project's GitHub repository; see Appendix A.

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Appendix A. Data, matching and variable definitions

Appendix A.1. Panel construction

PNAD Contínua draws a master sample of census tracts and interviews each selected household five times, once per quarter. Thirteen rotation groups are in the field over our sample period. Data collection is spread over the twelve weeks of the quarter.

Because no individual identifier is published, respondents have to be linked across interviews. The first pass of the stage-3 procedure is the classical rule of Ribas and Soares (2008), developed for the *IBGE* monthly employment survey (*PME*), which matches on the household identifier together with sex and date of birth while allowing for limited recording error. The two further passes described in Section 3 donate birth dates across a respondent's interviews and resolve fragmented sequences with a graph-theoretic match. Table A.1 reports the performance of the procedure as a whole.

Table A.1: Performance of the stage-3 panel identification

Panel A. Share linked across at least k interviews (%)

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
Individuals	100.0	83.3	70.6	59.2	46.0
Households	100.0	92.8	84.9	75.4	60.8

Panel B. Share matched into the next quarter, by interview (%)

	From 1st	From 2nd	From 3rd	From 4th	Overall
Employed workers	85.8	86.8	87.2	87.4	86.8

Notes: Computed on all quarterly *PNAD Contínua* rotation panels used in the paper. Panel A covers the whole panel population and reports the share of linked units observed in at least k of the five scheduled interviews. Panel B is restricted to workers employed at the interview date and reports the share the algorithm links into the following quarter, by the interview they are observed in; the fifth interview is the last scheduled one and is excluded. Stage 3 links respondents on household, sex and date of birth, donates birth dates across interviews, and resolves fragmented sequences with a graph-theoretic fuzzy match.

The build in this paper reconstructs the panel from the published quarterly microdata. Each quarter is downloaded once, cleaned with *Data Zoom*'s own routine and cached; the stage-3 identification is then run separately on each of the thirteen rotation groups. Because a household belongs to exactly one group and the algorithm only ever compares rows within the data it is given, running it group by group is equivalent to running it on the whole file, while keeping peak memory bounded. The sampling unit, the stratum and the interview counter survive the build as ordinary columns, which is what the inference and the attrition analysis need.

Appendix A.2. Variable definitions

In *PNAD Contínua* the working-age population is aged 14 and over. A person is employed if they worked at least one hour for pay in the reference week, worked without pay in a family business, or were temporarily absent from a job. A person is unemployed if not employed, having searched for work in the previous 30 days, and available to start.

Employment exit equals one when a person employed at t is unemployed or out of the labour force at $t+1$.

College degree equals one for workers who have completed tertiary education.

Formality follows the *PNAD Contínua* classification by employment category: private and public employees with a signed work card or statutory status, self-employed workers contributing to social security, and employers with a registered firm are formal; the remainder are informal.

Occupation comes from *IBGE*'s *Classificação de Ocupações para Pesquisas Domiciliares* (COD), the four-digit household-survey classification. Its leading digit is the major group, and we use those ten groups throughout. *Sector* is aggregated to five broad activity groups (agriculture, industry, construction, trade, services). Missing values in either are retained as an explicit "not reported" category rather than dropped, so that no observations are lost when they enter as fixed effects.

Tenure is the *PNAD Contínua* grouped time-in-current-job variable: under one month, one to eleven months, one to two years, two years or more.

Appendix A.3. Replication

The replication package lives in the project's GitHub repository. It contains the full *R* pipeline: `build/code` downloads the microdata from *IBGE* and rebuilds the panel, and `analysis/code` constructs the analysis sample, estimates every model and writes every table and figure in this paper. Random seeds, bootstrap replication counts, package versions and the *R* version are recorded in the package. The closed-form margins of equation (3) are checked there against a general average-predictions routine on a random subsample: the largest absolute discrepancy over the 52 quarters is of the order of 10^{-15}, and the check is written to `analysis/output/logs/08_margins_validation.txt`. The wild cluster bootstrap uses

9,999 replications and seed 20260615; the simultaneous bands use 10,000 multiplier draws; the decomposition bootstrap uses 500 replications. All computations were carried out in *R* (R Core Team, 2024) with the `fixest` package for high-dimensional fixed effects (Bergé, 2018).

Appendix B. Additional tables

Table B.1 reports the two pandemic contrasts under each of the alternative clustering schemes discussed in Section 4.3, and Table B.2 gives the composition of the estimation sample quarter by quarter.

Table B.1: Sensitivity of the college gap to the variance estimator

Clustering	Other quarters δ	Onset 2020Q1 $\delta + \omega_1$	Mid-pandemic $\delta + \omega_2$
Two-way: PSU and year-quarter	-0.010***	-0.035***	0.018***
	(0.001)	(0.001)	(0.002)
PSU	-0.010***	-0.035***	0.018***
	(0.000)	(0.003)	(0.001)
Household	-0.010***	-0.035***	0.018***
	(0.000)	(0.003)	(0.001)
Individual	-0.010***	-0.035***	0.018***
	(0.000)	(0.003)	(0.001)
Two-way: individual and year-quarter	-0.010***	-0.035***	0.018***
	(0.001)	(0.001)	(0.002)
Year-quarter	-0.010***	-0.035***	0.018***
	(0.001)	(0.001)	(0.002)

Notes: All rows report the same point estimates from the full specification, column (6) of Table 3; only the variance estimator changes. Number of clusters: 39,270 primary sampling units, 1,667,341 households, 3,024,050 individuals, 52 year-quarters. The onset contrast is a case in which no asymptotic cluster-robust variance is adequate: the deviation is identified from a single year-quarter cluster, and the cluster-robust variance is then known to be severely downward biased. The wild cluster bootstrap in Table 3 is the relevant benchmark for it.

Table B.2: Sample composition by quarter

Quarter t	Person-quarters	PSUs	College	Exit rate	Quarter t	Person-quarters	PSUs	College	Exit rate
2012Q1	163,486	11,937	0.142	0.097	2018Q3	166,945	12,070	0.197	0.092
2012Q2	170,903	11,943	0.142	0.092	2018Q4	166,219	12,059	0.200	0.104
2012Q3	170,330	11,944	0.141	0.092	2019Q1	164,214	12,048	0.204	0.094
2012Q4	169,612	11,978	0.142	0.103	2019Q2	167,449	12,051	0.205	0.095
2013Q1	170,738	12,008	0.147	0.098	2019Q3	166,586	12,023	0.202	0.091
2013Q2	171,660	11,991	0.145	0.098	2019Q4	156,885	11,957	0.208	0.106
2013Q3	173,793	12,040	0.148	0.093	2020Q1	126,921	11,452	0.228	0.128
2013Q4	174,805	12,031	0.149	0.101	2020Q2	95,785	10,760	0.245	0.095
2014Q1	175,337	12,028	0.154	0.091	2020Q3	91,990	11,006	0.251	0.053
2014Q2	178,121	12,055	0.156	0.088	2020Q4	84,865	10,662	0.247	0.065
2014Q3	179,089	12,026	0.153	0.086	2021Q1	81,186	10,671	0.248	0.056
2014Q4	178,704	12,082	0.161	0.100	2021Q2	97,659	11,220	0.239	0.047
2015Q1	177,773	12,054	0.166	0.093	2021Q3	118,529	11,678	0.223	0.050
2015Q2	178,148	12,046	0.167	0.090	2021Q4	128,843	11,873	0.223	0.119
2015Q3	174,128	12,050	0.171	0.103	2022Q1	133,987	11,890	0.226	0.098
2015Q4	172,894	12,030	0.171	0.110	2022Q2	144,244	12,017	0.221	0.097
2016Q1	170,818	12,067	0.181	0.108	2022Q3	145,338	12,002	0.228	0.092
2016Q2	170,502	12,068	0.178	0.109	2022Q4	142,361	11,951	0.227	0.102
2016Q3	167,652	12,061	0.180	0.102	2023Q1	139,129	11,937	0.231	0.091
2016Q4	167,302	12,055	0.190	0.113	2023Q2	143,014	11,926	0.232	0.090
2017Q1	164,368	12,029	0.184	0.098	2023Q3	145,785	11,994	0.232	0.091
2017Q2	166,563	12,069	0.186	0.098	2023Q4	147,396	12,014	0.236	0.101
2017Q3	166,530	12,058	0.187	0.095	2024Q1	148,343	12,008	0.238	0.088
2017Q4	166,738	12,049	0.187	0.105	2024Q2	151,072	12,005	0.240	0.084
2018Q1	163,878	12,052	0.197	0.099	2024Q3	149,665	12,019	0.238	0.085
2018Q2	164,360	12,030	0.197	0.091	2024Q4	149,486	12,060	0.239	0.097

Notes: Survey-weighted college share and employment-exit rate among workers employed in quarter t and matched to quarter $t+1$. The two blocks continue one another: the left block runs from 2012Q1 to 2018Q2 and the right from 2018Q3 to 2024Q4.